	Department of Software Engineering	Form: CAF
	School of Electrical Engineering and Computer Science (SEECS)	Date: 31-01-2024
	Corrective Action Request	Course: CS 212

Corrective Action Request for Spring 2024 Semester

Initiated by (Name)	Fahad Ahmed Satti	**Designation / Appointment**	Assistant Professor/ Knowledge Group Head – Programming Core
Nature of Problem:	Rationalization of CLOs and Unified content/programming language, for CS-212 Object Oriented Programming		

Assessment of Problem:

On January 30, 2024, the Programming Core Knowledge Group convened to assess and streamline the Course Learning Outcomes (CLOs) and their alignment with Program Learning Outcomes (PLOs) for CS-212, Object-Oriented Programming. During this session, the Knowledge Group decided to standardize the programming language and course content across the School of Electrical Engineering and Computer Science (SEECS) to ensure a consistent level of expertise among all students. Specifically, CS-212 will emphasize abstract delivery of Object-Oriented Programming (OOP) concepts, subsequently reinforcing these concepts through practical implementation using Java—a pure Object-Oriented Language.

Furthermore, the CLOs underwent a comprehensive review and updating process to accurately align with the appropriate Bloom's Taxonomy (BT) levels, thereby optimizing content delivery. The revised CLOs can be found in Annex A.

Faculty/DLQEC Name: Dr. Fahad Ahmed Satti	**Signature:**

Suggested corrective action(s):

- The centrally aligned outline should be used from Spring 2024.
- The updated CLOs should be adopted for all instances of CS-212.

Name (DLQEC): Mehwish Kiran	**Signature & Date: 01-02-2024**
Name (UG Program Coordinator): Dr. Bilal Ali	**Signature & Date: 01.02.2024**
Name (Head of the Department): Dr. Rafia Mumtaz	**Signature & Date: 01-02-2024**

Annex A

Table 1: Previous CLOs for CS 212 OOP

No.	Course Learning Outcomes	PLO (SE)	PLO (CS)	BT Level
CLO 1	**Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.	PLO-1	PLO-1	C-2
CLO 2	**Design** solutions to real-world problems by **applying** appropriate object-oriented techniques.	PLO-3	PLO-2	C-6
CLO 3	**Develop** programs using the latest IDEs as per standard practices applicable to the software industry.	PLO-5	PLO-5	P-3
CLO 4	**Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry.	PLO-8	PLO-8	A-3

Table 2: The proposed CLOs for CS 212 OOP

No.	Course Learning Outcomes	PLO	PLO	BT Level
		(SE)	(CS)	
CLO 1	**Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.	1 (Engineering Knowledge)	1 (Academic Education)	C-2
CLO 2	**Evaluate** complex programming problems using object-oriented principles analyzing solutions for efficiency and scalability	2 (Problem Analysis)	3 (Problem Analysis)	C-5
CLO 3	**Build** programs using the latest IDEs as per standard practices applicable to the software industry.	5 (Modern Tool Usage)	5 (Modern Tool Usage)	P-3
CLO 4	**Develop** solutions to real-world problems by applying appropriate object-oriented techniques.	3 (Design/Development of Solutions)	4 (Design/Development of Solutions)	C-6
CLO 5	**Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry.	PLO-8 (Ethics)	PLO-9 (Ethics)	A-3